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(54) **Title:** PRODUCING THE FZD7 CRD RECOMBINANT PROTEIN TO INHIBIT THE WNT SIGNALS AND CSCS WITH METHODS FOR IMPROVING THE SAME

(57) **Abstract:** The present invention focuses on creating a soluble FZD7 receptor as a potential treatment for cancer. This new method aims to inhibit the Wnt/[3-catenin signaling pathway, which is crucial in controlling cancer cell behaviors like growth and survival, especially in cancer stem cells. CSCs are difficult to treat due to their ability to spread, return, and resist common treatments, so blocking this signaling pathway could effectively target these tough cells. The soluble FZD7 receptor, produced through recombinant technology, binds to Wnt ligands that typically activate the Wnt/[β-catenin pathway. By preventing these ligands from linking to FZD7 cell surface receptors, the soluble receptor can hinder Wnt signaling, thus reducing the formation of CSCs. This invention can be applied to treat various cancers, including breast, gastrointestinal, lung, and skin cancers, potentially improving treatment effectiveness and minimizing side effects. This method represents a promising advance in targeted cancer therapies.

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Title of Invention: Producing The FZD7 CRD Recombinant Protein to Inhibit the Wnt Signals and CSCs with Methods for Improving the Same

[0001] Cancer stem cells (CSCs) play a fundamental role in the processes of metastasis, recurrence, tumor heterogeneity, resistance to chemotherapy and radiotherapy, and ultimately failure of cancer treatment. Abnormal activity of the Wnt signaling pathway is a key factor in forming and maintaining various CSCs, including in breast, gastrointestinal, lung, and skin cancers. Given the critical role of Wnt signaling in self-renewal, tumorigenesis, and differentiation of cancer cells into CSCs, targeting CSCs by inhibiting this pathway could be considered a promising therapeutic approach in the fight against cancer. The Wnt pathway includes ten receptors of the FZD family and nineteen Wnt ligands. Among these receptors, FZD7 is specifically overexpressed in many types of cancer, and this overexpression is directly associated with metastasis, recurrence, and drug resistance of cancer cells. In this invention, the first of its kind in the country, the extracellular domain of the FZD7 receptor (FZD7 CRD) has been recombinantly produced. In addition, the effects of FZD7 CRD in inhibiting proliferation and inducing apoptosis in AGS gastric cancer cells and SW480 colon cancer cells have been investigated. The soluble FZD7 CRD protein can inhibit the Wnt/β-catenin signaling pathway through competitive binding to Wnt ligands, thereby preventing the binding of these ligands to FZD7 receptors present on the cell surface. Therefore, the recombinant FZD7 CRD protein can effectively prevent the formation and maintenance of cancer stem cells by inhibiting the Wnt pathway and can be used as a novel therapeutic strategy in combating cancer.

[0002] As said, this invention includes the treatment of various types of cancers such as breast cancer, gastrointestinal cancer, lung cancer, and skin cancer. Due to the important role of the FZD7 receptor in tumor progression and metastasis, targeting this receptor could help improve the efficacy of anticancer therapies and reduce the likelihood of relapse and drug resistance. This innovation could greatly contribute to the development of targeted therapies and reduce the side effects associated with cancer treatments. It is also considered a promising candidate in the field of cancer treatment.

Technical Field

[0003] A61P – C12N 5/095 – A61K 48/00

Background Art

[0004] 1- Unlocking the Wnt pathway: Therapeutic potential of selective targeting FZD7 in cancer

[0005] March 2022

[0006] The Wnt signaling is of paramount pathophysiological importance. Despite showing promising anticancer activities in pre-clinical studies, current Wnt pathway inhibitors face complications in clinical trials resulting from on-target toxicity. Hence, the targeting of pathway component(s) that are essential for cancer but dispensable for normal physiology is key to the development of a safe Wnt signaling inhibitor. Frizzled7 (FZD7) is a Wnt pathway receptor that is redundant in healthy tissues but crucial in various cancers. FZD7 modulates diverse aspects of carcinogenesis, including cancer growth, metastasis, maintenance of cancer stem cells, and chemoresistance. In this review, we describe state-of-the-art knowledge of the functions of FZD7 in carcinogenesis and adult tissue homeostasis. Next, we overview the development of small molecules and biomolecules that target FZD7. Finally, we discuss challenges and possibilities in developing FZD7-selective antagonists.

[0007] The mentioned essay discusses the same subject we talked about in this application and the possibilities it offers for cancer treatment while the claimed patent described how a recombinant protein is produced to inhibit the Wnt signaling pathway and also ways to improve this method.

[0008] 2- CN106084067

[0009] Novel eukaryotic recombinant protein with tumor inhibition effect and preparation method thereof

[0010] The invention relates to a novel eukaryotic recombinant protein with a tumor inhibition effect and a preparation method thereof and belongs to the field of medical bioengineering. The novel eukaryotic recombinant protein with a tumor inhibition effect comprises 132 amino acids, 1th-25th amino acids form an NLS sequence, 26th-39th amino acids form a His-FLAG label sequence, 42th-121th amino acids form an N240 sequence, 122th-132th amino acids form a TAT sequence and the His-FLAG label sequence is bonded with the N240 sequence through two RS amino acids. The novel eukaryotic recombinant protein solves the problem that the existing small molecular drug cannot obstruct bonding of TCF4 and other cancer genes but beta-catenin so that the existing small molecular drug cannot effectively inhibit tumor stem cells, realizes comprehensive blocking of bonding of TCF4 and a protooncogene, effectively inhibits tumor stem cells and improves chemotherapeutic drug sensitivity.

[0011] The mentioned invention introduced a recombinant protein that is similar to the subject of our claimed one and both of them are efficient for treating cancers but their way of producing and also their target is different because our claimed protein inhibits the Wnt signaling pathway and stem cells while the mentioned patent aims for tumors.

[0012] 3- CN116712533

[0013] Application of SOSTDC1 protein in preparation of products for treating skin diseases

[0014] The invention provides application of SOSTDC1 protein in preparation of products for treating skin diseases, and belongs to the technical field of biological medicine. Wherein the SOSTDC1 protein comprises a recombinant SOSTDC1 protein, or a mutant of the recombinant SOSTDC1 protein, and a preparation method of the SOSTDC1 protein. The application of the SOSTDC1 protein in preparation of products for treating skin diseases provided by the invention proves that the SOSTDC1 recombinant protein can inhibit a Wnt signal channel of skin tissues, so that the keratinization of keratinocytes is reduced, the symptoms such as epidermal hyperplasia and desquamation are effectively blocked and delayed, and the skin diseases are effectively treated. The treatment and delay of skin diseases including psoriasis and skin fibrosis based on the SOSTDC1 protein are realized

[0015] The mentioned invention is close to our claimed one because both of them are proteins that inhibit the Wnt signal channel but the said patent is produced for skin disease and its SOSTDC1 protein while our claimed one is FZD7-CRD recombinant protein which aims for cancer stem cells and can be helpful for many cancer types.

[0016] 4- United States Patent 10913786

[0017] Compositions and methods for inhibiting Wnt signaling

[0018] The present disclosure relates to isolated polypeptides that inhibit Wnt signaling, pharmaceutical compositions comprising the isolated polypeptides, and methods of use thereof. Nucleic acids, cells, and methods of production related to the isolated polypeptides and compositions are also disclosed.

[0019] The mentioned patent relates to inhibiting Wnt signals which is the same as our claimed one but the said patent introduced some methods for this purpose while we described one FZD7-CRD protein and its improving ways.

[0020] 5- United States Patent Application 20190202909

[0021] ANTIBODIES AGAINST FRIZZLED RECEPTOR

[0022] This invention provides monoclonal antibodies that recognize one or more Frizzled receptors. The invention further provides methods of using such monoclonal antibodies as a therapeutic, diagnostic, and prophylactic.

[0023] The said invention uses monoclonal antibodies to fight against Wnt signals while our claimed patent produced a protein to inhibit these signals.

[0024] 6- US9987357B2

[0025] Methods and monitoring of treatment with a WNT pathway inhibitor

[0026] Methods for treating diseases such as cancer comprising administering a Wnt pathway inhibitor, either alone or in combination with other anti-cancer agents, and monitoring for skeletal-related side effects and/or toxicity.

[0027] The mentioned patent's goal is the same as our claimed one but it focuses more on the skeletal-related side of the treatment process and also the protein it mentions is FZD8 protein.

Summary of Invention

[0028] A recombinant protein called FZD7-CRD effectively inhibits the Wnt pathway, leading to reduced drug resistance and tumor growth in cancer stem cells. To prepare gene sequences, researchers retrieved reference sequences from the NCBI and UniProt databases and designed primers using bioinformatics software. The FZD7 gene, which has one exon, was amplified from human DNA through PCR, and the 474 bp band was inserted into the pTG19-T cloning vector. After verifying the sequence, the FZD7 gene was inserted into the pET28a+ expression vector and transferred to E. coli. The FZD7 protein was produced and purified using affinity chromatography. SDS-PAGE and Western blotting confirmed its expression. The FZD7 protein binds to Wnt ligands, preventing excessive activation of the Wnt pathway. If post-translational modifications are absent, mammalian expression systems can be used for production. Codons may be optimized to improve translation efficiency. Protein engineering techniques can enhance FZD7's stability and activity. Combining chromatography methods increases purity. Biophysical methods can evaluate FZD7 binding to Wnt ligands. 3D cell culture models help assess anticancer activity, and delivering FZD7 in nanoparticles can improve targeting. FZD7 may also be combined with chemotherapeutic drugs for better effects. Evaluating FZD7 in animal models and conducting clinical trials is essential for safety and efficacy. Molecular simulations may provide insights for new drugs, and FZD7 could enhance immunotherapy when combined with other treatments and gene editing techniques.

Technical Problem

[0029] More than 70% of cancer deaths occur in developing countries, and more than 60% of new cancer cases are reported in Africa, Asia, and Central and South America. Despite significant advances in surgery, chemotherapy, radiotherapy, and newer therapeutic approaches such as molecularly targeted therapies and immunotherapy, treatment outcomes remain unsatisfactory. One of the main challenges in this field is cancer stem cells (CSCs), which, due to their differentiation and self-renewal capabilities, are responsible for metastasis, cancer recurrence, and resistance to existing therapies. The Wnt signaling pathway plays a pivotal role as one of the key pathways in regulating cellular processes such as proliferation, differentiation, and self-renewal of stem cells, including CSCs. Overexpression or activation of this pathway leads to redifferentiation of cancer cells into CSCs, increased sensitivity to chemotherapeutic drugs, and ultimately increased tumorigenesis and resistance to treatment.

[0030] Wnt signaling is composed of a complex network of different receptors including ten Frizzled receptors (FZD), the tyrosine kinase receptors Ryk, ROR, and PTK7, the coreceptors LRP5/6, and nineteen Wnt ligands. Due to the complex and unpredictable interactions between these receptors and ligands, Wnt signaling is divided into three distinct pathways: the canonical β-catenin/TCF pathway, the planar cell polarity (PCP) pathway, and the Ca2+ pathway.

[0031] In adult tissues, Wnt signaling is normally silent, except in stem cells such as hematopoietic stem cells and intestinal crypts, which play important roles in tissue regeneration and repair. However, disruption of this pathway leads to uncontrolled cell proliferation and ultimately to cancer. Various types of cancer, including cancers with mutations in Wnt pathway components and cancers with increased expression of Wnt ligands and receptors, are dependent on this pathway.

[0032] FZD family receptors, particularly FZD7, have attracted attention due to their important role in cancer progression. Increased expression of FZD7 has been observed in many cancers, including colon, breast, liver, and lung cancers, and has been associated with tumor growth, invasion, and metastasis. In addition, FZD7 has been identified as a tumor biomarker in cancer stem cell populations, and its increased expression is associated with relapse and resistance to conventional therapies.

[0033] Therefore, targeting the FZD7 receptor as a therapeutic strategy to inhibit the Wnt signaling pathway could prevent the formation and survival of cancer stem cells and lead to reduced tumor invasion and metastasis. This invention aims to produce and purify a recombinant protein of the extracellular domain of the FZD7 receptor (FZD7-CRD), which can inhibit the formation of cancer stem cells, tumorigenesis, cancer cell invasion, and cancer recurrence by inhibiting the Wnt pathway.

[0034] Purposes of the invention:

[0035] The main objective of this invention is to develop a recombinant protein called FZD7-CRD that acts as an effective inhibitor of the Wnt pathway in the treatment of cancers dependent on this pathway. This protein inhibits the Wnt signaling pathway by binding to Wnt ligands and preventing the activation of the FZD7 receptor, thereby helping to reduce tumorigenesis, metastasis, and drug resistance in cancer cells. In particular, this recombinant protein can be used as a potential therapeutic tool to target cancer stem cells in various types of cancers and promises significant advances in the treatment of cancers that are resistant to current treatments.

Solution of Problem

[0036] Despite extensive advances in cancer prevention, diagnosis, and treatment, cancer remains a leading cause of death worldwide. One of the major challenges in cancer treatment is drug resistance and tumor recurrence, which is mainly driven by CSCs. These cells are difficult to target by conventional therapies such as chemotherapy and

radiotherapy due to their unique properties, such as their ability to be arrested in the G0 phase of the cell cycle. Therefore, the development of new therapies that directly target CSCs is urgently needed. In this context, the Wnt signaling pathway plays a critical role in regulating the survival and proliferation of CSCs. The Wnt pathway plays an important role in the survival of cancer cells and their ability to metastasize by activating several proteins, including Frizzled receptors, especially FZD7. Research has shown that abnormal expression of the FZD7 receptor is increased in various types of cancer, such as colon, stomach, liver, and breast cancer.

[0037] Innovation in the production of FZD7 protein:

[0038] In this invention, the extracellular part of the FZD7 receptor was produced recombinantly in a prokaryotic system. The production of this protein in prokaryotic systems poses certain challenges; including problems with protein stability, correct folding, and its effective purification. To overcome these challenges, affinity chromatography (IMAC) was used to purify recombinant proteins with histidine sequences. This method not only has high efficiency in protein purification but also allows its use in applied and therapeutic studies by preserving the natural structure of the protein.

[0039] The FZD7 protein produced in this study can prevent excessive activation of the Wnt pathway by binding to Wnt ligands and preventing their binding to membrane FZD7 receptors. Thus, soluble FZD7 as a recombinant protein can be used as a targeted drug to treat cancers in the future.

[0040] Details of the gene sequence preparation process:

[0041] The reference nucleotide and amino acid sequences of the FZD7 receptor were retrieved from the NCBI and UniProt databases, and specific primers were designed for these sequences using various bioinformatics software, such as CLC and AlleleID. The FZD7 gene, which contains only one exon, was amplified from human DNA by polymerase chain reaction (PCR). The PCR reaction was performed in a volume of 60 µl to extract the 474 bp band from the gel, and then this gene was inserted into the pTG19-T cloning vector.

[0042] Finally, after confirming the sequence and correctness of cloning, the FZD7 gene was inserted into the pET28a+ expression vector and transferred to the prokaryotic system E. coli strain Shuffle. This strain was selected due to its unique ability to fold and express recombinant proteins correctly.

[0043] Protein examination and confirmation:

[0044] SDS-PAGE and Western blotting confirmed the successful expression of the FZD7 protein. In these experiments, the expressed proteins were electrophoresed alongside a molecular weight marker, and the presence and structural integrity of the recombinant protein were confirmed using specific antibodies against FZD7. In addition, protein

purification was performed using the affinity chromatography method (IMAC), which allows for rapid and efficient protein purification due to the presence of a histidine sequence.

[0045] This application demonstrates the feasibility of successfully producing recombinant FZD7 protein in a prokaryotic system and purifying it with high quality. Given the critical role of FZD7 in cancer cell survival and metastasis, this protein can be used as a promising candidate in the development of targeted drugs for the treatment of various types of cancers. Future research in this field could include preclinical and clinical trials to investigate the efficacy and safety of this recombinant drug.

[0046] To further improve and develop the method of production and use of recombinant FZD7 protein in cancer therapy, several other specialized solutions can be considered. These solutions include optimizing the various steps of protein production, developing advanced methods for assessing protein function, as well as exploring new clinical applications. These solutions can lead to cost reduction, increased efficiency, and development of more effective therapies:

[0047] 1- Optimization of protein expression system in eukaryotic cells:

[0048] • Use of mammalian expression system: Although prokaryotic expression systems are cost-effective for producing recombinant proteins, the produced proteins may lack post-translational modifications (such as glycosylation) that are essential for the biological activity of some proteins. The use of mammalian expression systems such as HEK293 or CHO can help produce FZD7 protein with more optimized structure and activity.

[0049] • Codon optimization: Using bioinformatics software, the codons of the FZD7 gene can be optimized to increase translation efficiency in selected host cells. This will lead to increased production of the recombinant protein.

[0050] 2- Engineering the FZD7 protein to increase stability and biological activity:

[0051] • Engineered changes in the protein sequence: Using protein engineering techniques, point mutations or small deletions and additions can be made in the FZD7 protein sequence to increase its stability and activity.

[0052] • Addition of protective peptide sequences: Adding peptide sequences such as TAT or PTD can help increase the transport of the FZD7 protein into cells and prevent its rapid degradation by proteases.

[0053] 3- Use of modern technologies in protein purification:

[0054] • Multidimensional chromatography: Using a combination of affinity chromatography and ion exchange chromatography can lead to increased purity and quality of the FZD7 protein.

[0055] • Size-exclusion chromatography: This method can separate recombinant proteins of different sizes and remove oligomers and unwanted protein aggregates.

[0056] 4- Using advanced methods to evaluate protein function:

[0057] • Using biophysical methods: To evaluate the binding of FZD7 protein to Wnt ligands, biophysical methods such as SPR (Surface Plasmon Resonance) or ITC (Isothermal Titration Calorimetry) can be used. These methods provide detailed information about the binding constants and interaction dynamics.

[0058] • 3D cell models: To more accurately evaluate the anticancer activity of FZD7 protein, 3D cell culture models such as spheroids and organoids can be used, which better represent cell-cell and tumor interactions in vivo.

[0059] 5- Development of innovative drug formulations:

[0060] • Use of nanoparticles: Loading FZD7 protein into polymeric or lipid nanoparticles can help increase stability and better targeting of cancer tumors.

[0061] • Combination with other drugs: FZD7 protein can be combined with chemotherapeutic drugs or inhibitors of other pathways such as PI3K/Akt or MAPK to provide enhanced therapeutic effects.

[0062] 6- Comprehensive preclinical and clinical studies:

[0063] • Evaluation in animal models: In addition to in vitro experiments, evaluation of FZD7 protein in animal models of various cancers, including evaluation of metastasis and survival, is essential to confirm efficacy and safety.

[0064] • Initiation of clinical trials: After successful results in preclinical studies, it will be necessary to conduct phase 1 and 2 clinical trials to evaluate the safety, optimal dosage, and efficacy of FZD7 protein in cancer patients.

[0065] 7- Use of bioinformatics and molecular modeling:

[0066] • Molecular dynamics simulations: Using molecular dynamics simulations, the structure and stability of the FZD7 protein and its interactions with Wnt ligands can be studied at the atomic level. This information can help optimize the design of new proteins and drugs.

[0067] • Identify new targets: Bioinformatics techniques are used to identify other key components of the Wnt pathway that may be used as potential drug targets in cancer treatment.

[0068] 8- Explore the potential for use in combination therapies:

[0069] • Use of FZD7 in immunotherapy: FZD7 protein can be used as part of an immunotherapy strategy, for example in combination with checkpoint inhibitors, to enhance the antitumor immune response.

[0070] • Combination with gene therapies: The use of gene editing techniques such as CRISPR-Cas9 to modify key Wnt pathway genes, combined with the use of the FZD7 protein, could lead to novel combination therapies that simultaneously target multiple aspects of cancer development.

[0071] These solutions could help further develop and improve the initial invention, creating more effective and specific cancer therapies.

Advantage Effects of the Invention

[0072] 1- Specific inhibition of the binding of Wnt ligands to the FZD7 receptor: FZD7 CRD directly interacts with the FZD7 receptor at the cellular level and prevents the binding of Wnt ligands to this receptor. This specific inhibition inactivates the Wnt signaling pathway and prevents the activation of processes that lead to the growth and survival of cancer stem cells.

[0073] 2- Self-protein with minimal stimulation of immune responses: Since FZD7 CRD is a self-protein, the likelihood of stimulating harmful immune responses is very low. This increases safety and reduces side effects caused by unwanted immune responses that usually occur with the use of foreign biological drugs.

[0074] 3- Reduction of cancer stem cells, migration, and metastasis: Inhibition of the Wnt pathway using FZD7 CRD leads to a significant reduction in the number of cancer stem cells, which are directly related to metastasis and cancer recurrence. This can help reduce tumor invasion and metastasis, and thus, contribute to better disease control.

[0075] 4- Reduction of side effects caused by chemotherapy due to specific action: Since FZD7 CRD specifically targets the Wnt pathway, side effects associated with chemotherapy that act non-specifically are reduced. This reduction in side effects leads to an improvement in the quality of life of patients.

[0076] 5- Increased patient survival: By reducing the abundance of cancer stem cells and inhibiting metastasis, the use of FZD7 CRD can lead to an increase in the survival of cancer patients. This is of particular importance, especially for patients facing treatment-resistant types of cancer.

[0077] 6- Reduction in treatment costs due to the use of chemical drugs: Due to the specific function of FZD7 CRD and the reduction in the need to use expensive and complicated chemical drugs, the treatment costs of patients are reduced. This is cost-effective for both patients and health systems.

[0078] 7- Domestic production capability: The technology for producing FZD7 CRD exists within the country and can be produced with existing facilities. This contributes to self-sufficiency in drug production and reduces dependence on the import of biological products.

[0079] 8- Lower production cost compared to monoclonal antibodies: The production of FZD7 CRD is cheaper than other biological drugs, including monoclonal antibodies. This economic advantage can help patients access this treatment better and reduce the overall costs of cancer treatment.

[0080] 9- The high potential of FZD7 CRD as a new and effective treatment method in combating cancer which can help improve treatment outcomes and reduce treatment-related costs.

Brief Description of Drawings

[0081] [Fig.1] Shows the steps taken in the synthesis and functional validation of the recombinant CRD FZD7 protein.

Description of Embodiments

[0082] [Fig.1] Shows the steps taken in the synthesis and functional validation of the recombinant CRD FZD7 protein, the first step is the amplification of the FZD7 extracellular domain gene sequence by polymerase chain reaction and gel purification, the second step is the insertion of the FZD7 fragment into the T-vector plasmid (pTG19) and generation of the PTG19-T-Fzd7 construct. The third step is the transformation of the PTG19-T-FZD7 construct into E. coli strain TOP-10. The fourth level is the confirmation of the FZD7 fragment in the plasmid vector by PCR colony, sequencing, and restriction endonuclease digestion methods. The fifth level is the subcloning of the nucleotide sequence of the extracellular domain of the FZD7 receptor in the pET28a+ expression plasmid. For the sixth step, the transformation of the FZD7-pET28a+ construct into E. coli strain Shuffle will be done. The seventh step is the induction of recombinant protein expression, then the confirmation of protein expression by SDS-PAGE and Western blot methods will be done for the eighth part. The ninth step is the purification of recombinant protein using combined affinity chromatography using the Ni-NTA column. The final steps are the confirmation of the inhibitory role of recombinant CRD FZD7 protein in inhibiting proliferation in gastric and colon cancer cell lines using the MTT assay and the confirmation of the inhibitory role of recombinant CRD FZD7 protein in inducing apoptosis in gastric and colon cancer cell lines using the Annexin/PI method.

Examples

[0083] Methodology for applying the FZD7 CRD invention in cancer treatment:

[0084] - Phase 3: Compare the FZD7 CRD protein with existing standard treatments in a large group of patients to confirm efficacy and safety.

[0085] • Finally, if the results are successful, FZD7 CRD will be approved by regulatory authorities as a new drug for the treatment of cancer.

[0086] Phase 4: Commercialization and Industrial-Scale Production:

[0087] • After clinical approval, industrial-scale production of FZD7 CRD will begin. This phase includes optimizing the manufacturing, packaging, and distribution processes for the product to healthcare facilities.

[0088] • Educational programs will be implemented for physicians and specialists to ensure optimal use of this drug in the treatment of cancer patients.

[0089] This implementation process includes all the steps necessary to move a product from the research phase to clinical treatment and ultimately to the market, and given the complexity of the processes, special attention to quality control and monitoring of each step is necessary to ensure the effectiveness and final safety of the product.

[0090] [Pic. 1] shows the nucleotide sequence of the amplified FZD7 fragment:

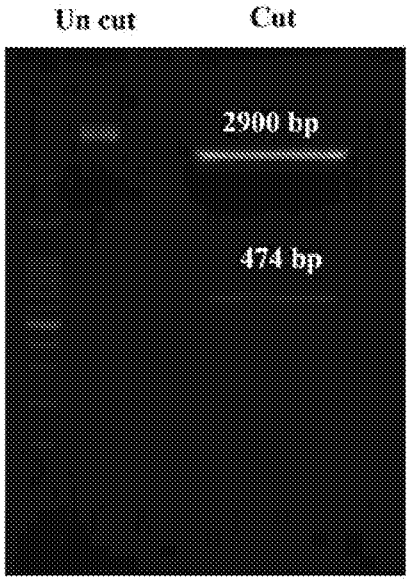
[0091]

ggatccCAGCCGTACCACGGAGAGAAGGGCATCTCCGTGCCGGACCACGGCTTCTGCCAGCCCATCTCCAT
CCCGCTGTGCACGGACATCGCCTACAACCAGACCATCCTGCCCAACCTGCTGGGCCACACGAACCAAG
AGGACGCGGGCCTCGAGGTGCACCAGTTCTACCCGCTGGTGAAGGTGCAGTGTTCTCCCGAACTCCGC
TTTTTCTTATGCTCCATGTAIGCGCCCGTGTGCACCGTGCTCGATCAGGCCATCCCGCCGTGTCGTTCTCT
GTGCGAGCCGCCCCGCCAGGGCTGCGAGGCGCTCATGAACAAGTTCGGCTTCCAGTGGCCCGAGCGGC
TGCGCTGCGAGAACTTCCCGGTGCACGGTGCGGGCGAGAICTGCGTGGGCCAGAACACGTCGGACGGC
TCCGGGGGCCCAGGCGGCGGCCCCACTGCCTACCCTACCGCGCCCTACCTGggatcc

[0092] Amplification of the FZD7 gene by a forward primer containing the BamH1 endonuclease cleavage site and a reverse primer containing the EcoRI endonuclease cleavage site

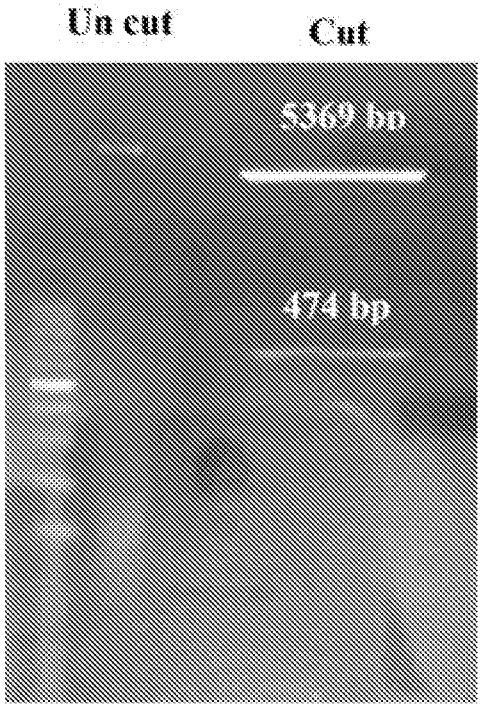
[0093] [Pic .2] shows the verification of the PTG19-T FZD7 construct by digestion with BamHI and EcoRI enzymes and observation of fragments of 474 and 2900 nucleotides. After amplification of the FZD7 gene by a forward primer containing a BamH1 endonuclease cleavage site and a reverse primer containing an EcoRI endonuclease cleavage site, the amplified fragment was transferred into the PTG19-T cloning plasmid. The nucleotide sequence of the amplified FZD7 fragment is shown in [Fig.1]. The PTG19-T FZD7 construct was verified by digestion with BamHI and EcoRI enzymes and observation of pieces of 474 and 2900 nucleotides:

[0094]



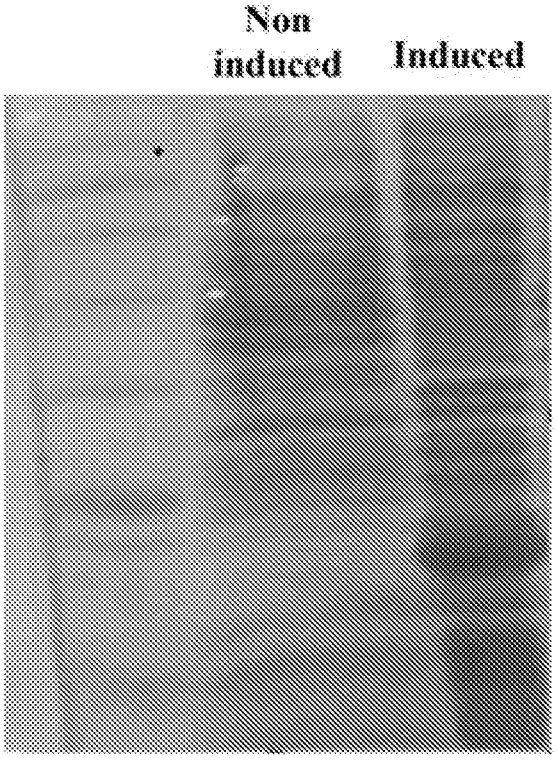
[0095] [Pic. 3] Verification of the pET28a+- FZD7 construct with BamHI and EcoRI endonucleases. The PTG19-T FZD7 construct was obtained by digestion with BamHI and EcoRI enzymes and fragments of 474 and 5369 nucleotides were observed:

[0096]



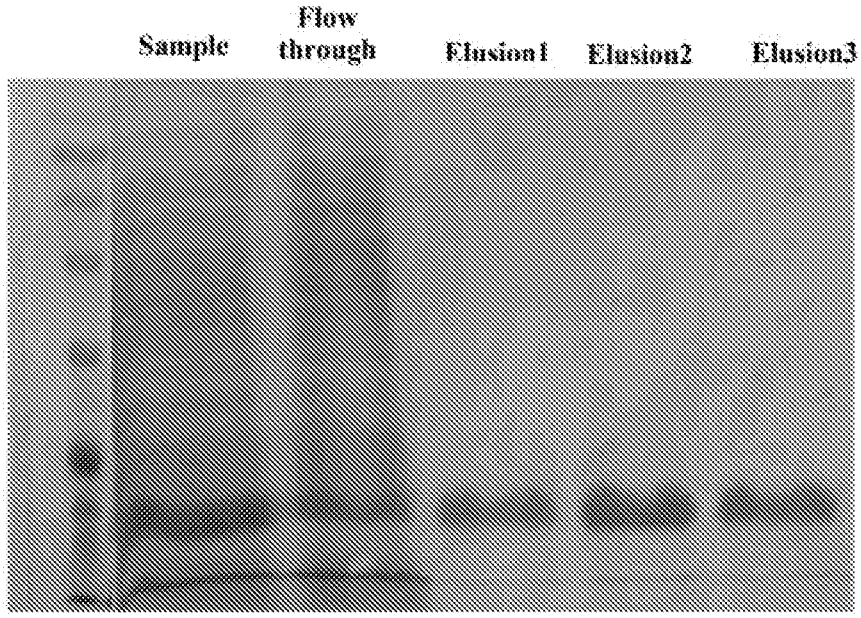
[0097] [Pic. 4] 20 kDa band of the extracellular portion of FZD7 protein after induction of expression in Shuffle bacteria. After the transformation of FZD7 CRD pET28a+- into Escherichia coli expression bacteria Shuffle T7 strain, expression induction was performed at OD6 with 0.5 mM IPTG. SDS-PAGE result shows expression of FZD7 CRD protein in the induced sample:

[0098]



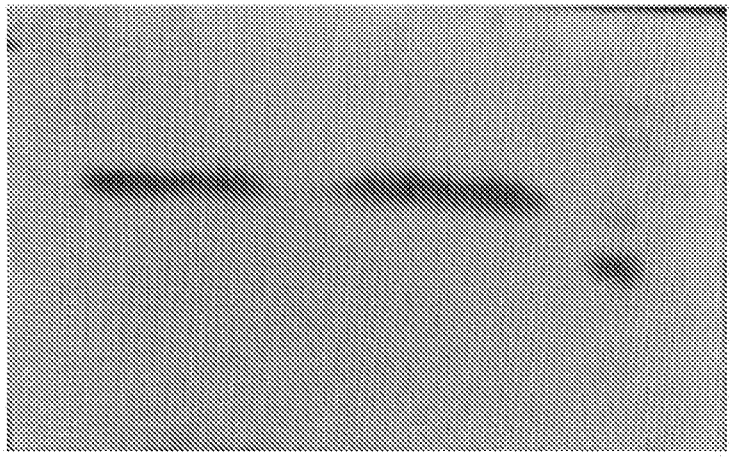
[0099] [Pic. 5] Describes 20 kDa bands of the FZD7 CRD protein after purification by affinity chromatography. The FZD7 CRD protein has a histidine sequence at its amino terminus, and purification of this protein was performed by affinity chromatography:

[0100]



[0101] [Pic. 6] Shows confirmation of the CRD protein of FZD7 by Western blot. In the Western blot, an antibody specific to the CRD domain of the FZD7 protein was used. The Western blot result shows the production of soluble CRD protein of FZD7:

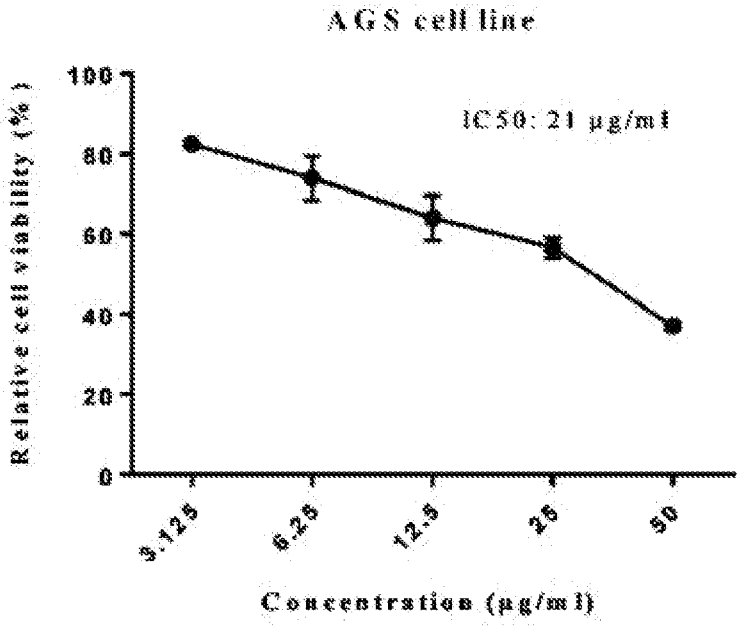
[0102]



[0103] The effect of FZD7 CRD on the proliferation rate of AGS gastric cancer cell line at concentrations of 50 µg/ml, 25 µg/ml, 12.5 µg/ml, 6.25 µg/ml, 3.12 µg/ml, and 1.56 µg/ml was measured by MTT assay after 72 hours of treatment.

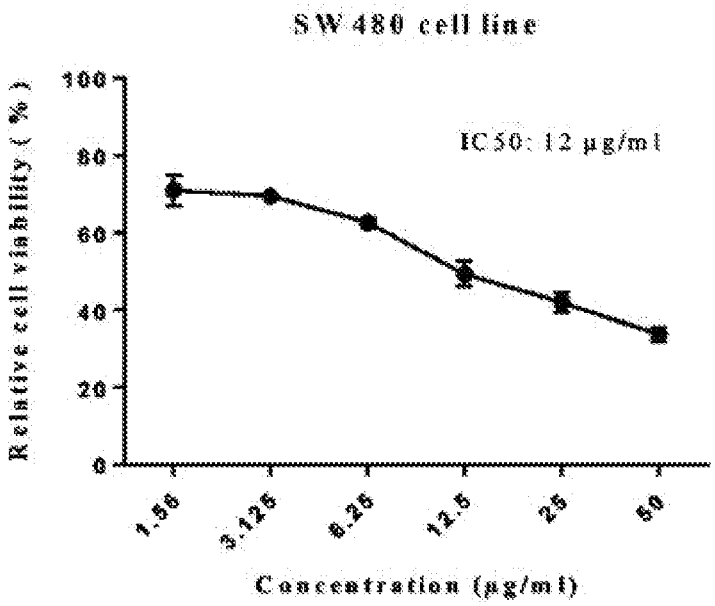
[0104] [Chart. 1] The inhibitory effect of FZD7 CRD on the proliferation of AGS gastric cancer cell lines was demonstrated:

[0105]



[0106] [Chart. 2] The inhibitory effect of FZD7 CRD is shown in inhibiting the proliferation of SW480 colon cancer cell lines:

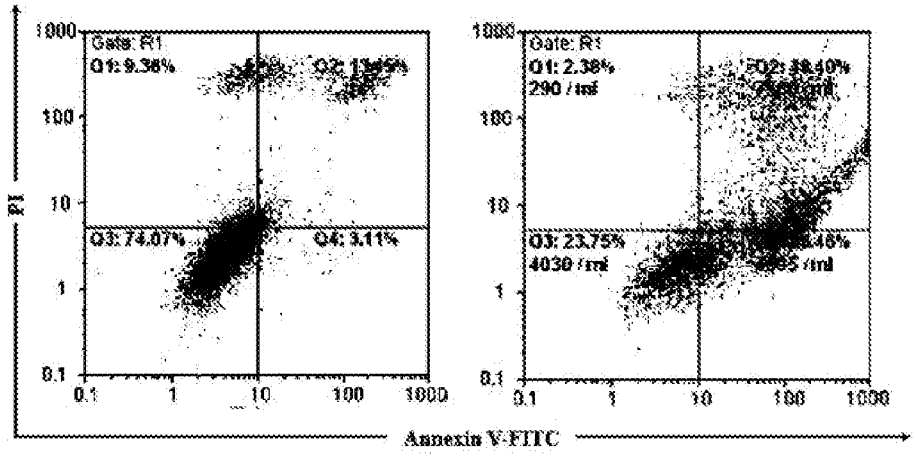
[0107]



[0108] The effect of FZD7 CRD on the proliferation rate of SW480 colon cancer cell line at concentrations of 50 µg/ml, 25 µg/ml, 12.5 µg/ml, 6.25 µg/ml, 3.12 µg/ml, and 1.56 µg/ml was measured by MTT method after 72 hours of treatment.

[0109] [Chart. 3] The inhibitory effect of FZD7 CRD on apoptosis induction of AGS gastric cancer cell lines is shown in this image:

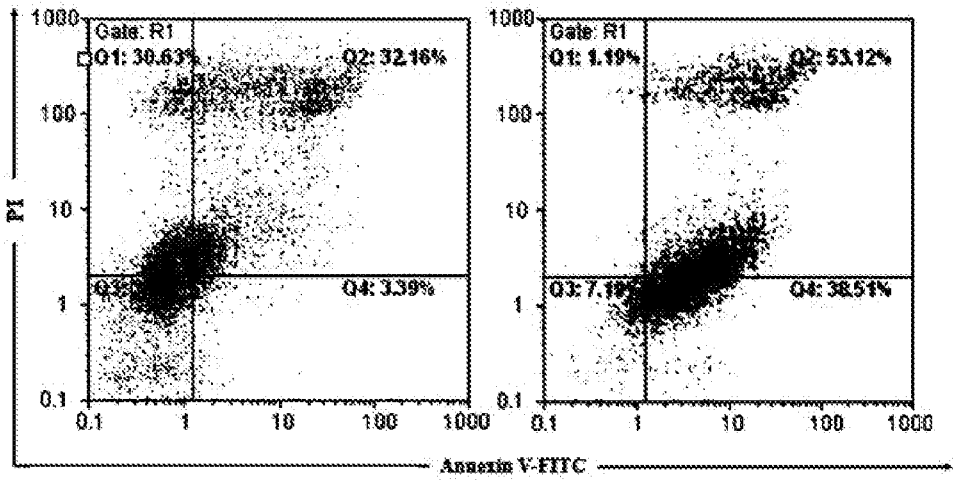
[0110]



[0111] The effect of FZD7 CRD on apoptosis induction of AGS gastric cancer cell line at a concentration of 25 µg/ml was measured by the Annexin/PI method after 72 hours of treatment.

[0112] [Chart. 4] The inhibitory effect of FZD7 CRD on apoptosis induction of SW480 colon cancer cell lines is shown in this image:

[0113]



[0114] The effect of FZD7 CRD on apoptosis induction of the SW480 gastric cancer cell line at a concentration of 20 µg/ml was measured after 72 hours of treatment by the Annexin/PI method.

Industrial Applicability

[0115] Recombinant FZD7 CRD protein is beneficial in the field of medical science and cancer treatment. Currently, recombinant proteins are one of the growing and vital sectors in the biotechnology and pharmaceutical industries, accounting for about 10% of the global pharmaceutical market. These proteins play a fundamental role in treating various diseases, including cancer, and are considered suitable alternatives to traditional treatment methods.

[0116] Industrial applications of FZD7 CRD protein:

[0117] 1. Cancer treatment with control and reduction of cancer stem cells: FZD7 CRD protein is specifically designed to inhibit the Wnt-FZD7 signaling pathway. This pathway plays a key role in the formation and survival of cancer stem cells. By targeting and inhibiting this pathway, FZD7 CRD protein can effectively control and reduce cancer stem cells, which helps to reduce cancer invasion and metastasis.

[0118] - Reducing cancer recurrence and metastasis:

[0119] FZD7 CRD protein can help reduce the occurrence of disease recurrence and metastasis in cancer patients, which is especially important for patients who do not respond to chemotherapy.

[0120] 2. Preventing chemotherapy side effects:

[0121] Reducing dependence on chemical drugs: Due to the specific targeting of the FZD7 CRD protein, it is possible to reduce the side effects caused by chemical drugs and improve the quality of life of patients.

[0122] 3. Reducing and optimizing treatment costs:

[0123] FZD7 CRD protein can help reduce the cost of cancer treatment and the need for expensive chemical drugs. Especially considering that it is less expensive to produce

than other biological drugs such as monoclonal antibodies, this protein can be a cost-effective option for cancer treatment.

[0124] 4. Domestic Commercialization and Production:

[0125] Given the existence of FZD7 CRD protein production technology in the country, this invention has the potential to be produced and commercialized domestically. This can help strengthen the domestic biotechnology industry and reduce dependence on foreign drug imports.

[0126] Overall Impacts:

[0127] With the confirmation of FZD7 CRD protein in additional laboratory tests and clinical studies, this drug can be mass-produced and commercialized, which in turn will greatly help reduce the costs of cancer treatment and chemotherapy drugs. Given the high prevalence of cancer and the need for effective and low-complication treatments, FZD7 CRD can play an important role in improving treatment outcomes and quality of life for cancer patients.

Claims

[Claim 1] A recombinant protein called FZD7-CRD participates as an inhibitor in the Wnt signaling pathway, leading to reduced drug resistance, metastasis, and tumorigenesis in cancer stem cells, and methods for developing the same.

[Claim 2] According to claim 1, to prepare gene sequences, reference nucleotide and amino acid sequences of the FZD7 receptor were retrieved from the NCBI and UniProt databases, and specific primers were designed for these sequences using various bioinformatics software, such as CLC and AlleleID.

[Claim 3] According to claim 2, the FZD7 gene, which contains only one exon, was amplified from human DNA by polymerase chain reaction (PCR), and then the PCR reaction was performed in a volume of 60 μl to extract the 474 bp band from the gel to insert this gene into the pTG19-T cloning vector.

[Claim 4] According to claim 3, after confirming the sequence and correctness of cloning, the FZD7 gene was inserted into the pET28a+ expression vector and transferred to the prokaryotic system of the E. coli strain Shuffle.

[Claim 5] According to claim 4, the extracellular portion of the FZD7 receptor was produced recombinantly in a prokaryotic system and the affinity chromatography (IMAC) method was used to purify recombinant proteins with histidine sequences, purify and maintain the protein structure.

[Claim 6] According to claim 5, the expression of the FZD7 protein was confirmed by SDS-PAGE and Western blotting methods by placing it next to a molecular weight marker for electrophoresis and using specific antibodies against FZD7.

[Claim 7] According to claim 6, the produced FZD7 protein solution, by binding to Wnt ligands, prevents their binding to FZD7 membrane receptors and excessive activation of the Wnt pathway.

[Claim 8] According to claim 1, if the produced proteins lack post-translational modifications (such as glycosylation), the use of a mammalian expression system such as HEK293 or CHO can help produce the FZD7 protein.

[Claim 9] According to claim 1, using bioinformatics software, the codons of the FZD7 gene can be optimized to increase translation

efficiency in selected host cells, which also increases the production of the recombinant protein.

[Claim 10] According to claim 1, utilizing protein engineering techniques, point mutations or small deletions and additions can be made in the FZD7 protein sequence to increase its stability and activity.

[Claim 11] According to claim 1, peptide sequences such as TAT or PTD can help increase the transport of the FZD7 protein into cells and prevent its rapid degradation by proteases.

[Claim 12] According to claim 1, size-exclusion chromatography is used to separate recombinant proteins of different sizes and remove oligomers plus unwanted protein aggregates.

[Claim 13] According to claim 1, to assess the binding of FZD7 protein to Wnt ligands, biophysical methods such as SPR (Surface Plasmon Resonance) or ITC (Isothermal Titration Calorimetry) can be used to obtain detailed information about the binding constants and interaction dynamics.

[Claim 14] According to claim 1, to more accurately assess the anticancer activity of the FZD7 protein, three-dimensional cell culture models such as spheroids and organoids can be used, which better represent cell-cell and tumor interactions in vivo.

[Claim 15] According to claim 1, loading FZD7 protein into polymeric or lipid nanoparticles can increase the stability and targeting of cancer tumors.

[Claim 16] According to claim 1, the FZD7 protein can be combined with chemotherapeutic drugs or inhibitors of other pathways such as PI3K/Akt or MAPK to provide enhanced therapeutic effects.

[Claim 17] According to claim 1, using molecular dynamics simulations, the structure and stability of the FZD7 protein and its interactions with Wnt ligands can be studied at the atomic level to provide a basis for the design of new proteins and drugs.

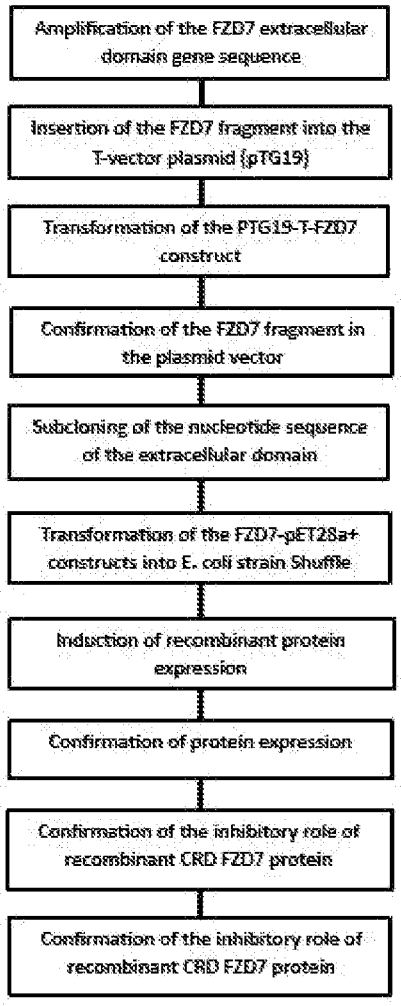
[Claim 18] According to claim 1, bioinformatics techniques are used to identify other key components of the Wnt pathway.

[Claim 19] According to claim 1, the FZD7 protein can be used as part of an immunotherapeutic strategy, for example in combination with checkpoint inhibitors, to enhance the antitumor immune response.

[Claim 20] According to claim 1, gene editing techniques such as CRISPR-Cas9 to make modifications to key Wnt pathway genes, combined with the use of the FZD7 protein, could lead to new combination

therapies that simultaneously target multiple aspects of cancer development.

[Fig. 1]


Amplification of the FZD7 extracellular domain gene sequence
Insertion of the FZD7 fragment into the T-vector plasmid (pTG19)
Transformation of the PTG19-T-FZD7 construct
Confirmation of the FZD7 fragment in the plasmid vector
Subcloning of the nucleotide sequence of the extracellular domain
Transformation of the FZD7-pET28a+ constructs into E. coli strain Shuffle
Induction of recombinant protein expression
Confirmation of protein expression
Confirmation of the inhibitory role of recombinant CRD FZD7 protein
Confirmation of the inhibitory role of recombinant CRD FZD7 protein

[Fig. 1]

INTERNATIONAL SEARCH REPORT

International application No.
PCT/IB2025/052752

A. CLASSIFICATION OF SUBJECT MATTER
A61K38/17,C07K14/705,A61P35/00 Version=2025.01

According to International Patent Classification (IPC) or to both national classification and IPC

B. FIELDS SEARCHED

Minimum documentation searched (classification system followed by classification symbols)

A61K, C07K, A61P

Documentation searched other than minimum documentation to the extent that such documents are included in the fields searched

Electronic database consulted during the international search (name of database and, where practicable, search terms used)

PatSeer, IPO Internal Database

C. DOCUMENTS CONSIDERED TO BE RELEVANT

Category*	Citation of document, with indication, where appropriate, of the relevant passages	Relevant to claim No.
X	Sloan, T. L. (2011). Characterization of Receptor Mediated Interactions with Wnt Proteins [Master's thesis, University of California, San Diego]. Whole document	1-20

☐ Further documents are listed in the continuation of Box C. ☐ See patent family annex.

* Special categories of cited documents:
"A" document defining the general state of the art which is not considered to be of particular relevance
"D" document cited by the applicant in the international application
"E" earlier application or patent but published on or after the international filing date
"L" document which may throw doubts on priority claim(s) or which is cited to establish the publication date of another citation or other special reason (as specified)
"O" document referring to an oral disclosure, use, exhibition or other means
"P" document published prior to the international filing date but later than the priority date claimed
"T" later document published after the international filing date or priority date and not in conflict with the application but cited to understand the principle or theory underlying the invention
"X" document of particular relevance; the claimed invention cannot be considered novel or cannot be considered to involve an inventive step when the document is taken alone
"Y" document of particular relevance; the claimed invention cannot be considered to involve an inventive step when the document is combined with one or more other such documents, such combination being obvious to a person skilled in the art
"&" document member of the same patent family

Date of the actual completion of the international search	Date of mailing of the international search report
09-07-2025	09-07-2025
Name and mailing address of the ISA/ Indian Patent Office Plot No.32, Sector 14,Dwarka,New Delhi-110075 Facsimile No.	Authorized officer P Jyothish Kumar Telephone No. +91-1125300200